MongoDB Aggregation Assignment – E-Commerce Analytics & Reporting System

REPORT 1: Monthly Sales Report

MongoDB Aggregation Query:

```
db.orders.aggregate([
 { $match: { status: "Delivered" } },
 { $group: {
    _id: { year: { $year: "$orderDate" }, month: { $month: "$orderDate" } },
    totalRevenue: { $sum: "$totalAmount" },
    totalOrders: { $sum: 1 },
    avgOrderValue: { $avg: "$totalAmount" }
 }},
 { $project: {
    year: "$_id.year",
    month: "$_id.month",
    totalRevenue: 1,
    totalOrders: 1,
    avgOrderValue: 1,
    _id: 0
 }},
 { $sort: { totalRevenue: -1 } }
]);
```

REPORT 2: Top 5 Selling Products

MongoDB Aggregation Query:

```
db.orders.aggregate([
 { $unwind: "$items" },
 { $lookup: {
    from: "products",
    localField: "items.productId",
    foreignField: "_id",
```

```
  as: "product"
}},
{ $unwind: "$product" },
{ $group: {
  _id: "$items.productId",
  productName: { $first: "$product.name" },
  category: { $first: "$product.category" },
  brand: { $first: "$product.brand" },
  totalQuantitySold: { $sum: "$items.quantity" },
  totalRevenue: { $sum: { $multiply: ["$items.quantity", "$items.price"] } }
}},
{ $sort: { totalQuantitySold: -1 } },
{ $limit: 5 }
]);
```

REPORT 3: Revenue by State

MongoDB Aggregation Query:

```
db.orders.aggregate([
{ $lookup: {
  from: "users",
  localField: "userId",
  foreignField: "_id",
  as: "user"
}},
{ $unwind: "$user" },
{ $group: {
  _id: "$user.state",
  totalRevenue: { $sum: "$totalAmount" },
  customers: { $addToSet: "$userId" }
}},
{ $project: {
  state: "$_id",
  totalRevenue: 1,
  totalCustomers: { $size: "$customers" },
  avgSpendingPerCustomer: { $divide: ["$totalRevenue", { $size: "$customers" }] },
  _id: 0
}}
]);
```

REPORT 4: Premium vs Non-Premium Customer Analysis

MongoDB Aggregation Query:

```
db.orders.aggregate([
 { $lookup: {
    from: "users",
    localField: "userId",
    foreignField: "_id",
    as: "user"
 }},
 { $unwind: "$user" },
 { $group: {
    _id: "$user.isPremium",
    totalRevenue: { $sum: "$totalAmount" },
    avgOrderValue: { $avg: "$totalAmount" }
 }},
 { $project: {
    userType: { $cond: [{ $eq: ["$_id", true] }, "Premium", "Non-Premium"] },
    totalRevenue: 1,
    avgOrderValue: 1,
    _id: 0
 }}
]);
```

REPORT 5: Product Rating Performance

MongoDB Aggregation Query:

```
db.products.aggregate([
 { $lookup: {
    from: "reviews",
    localField: "_id",
    foreignField: "productId",
    as: "reviews"
 }},
 { $project: {
    name: 1,
    avgRating: { $ifNull: [{ $avg: "$reviews.rating" }, 0] },
    totalReviews: { $size: "$reviews" }
 }},
 { $match: {
```

```
    $or: [
     { avgRating: { $lt: 3 } },
     { totalReviews: 0 }
    ]
 }}
]);
```

REPORT 6: Payment Mode Analytics

MongoDB Aggregation Query:

```
db.orders.aggregate([
 { $group: {
    _id: "$paymentMode",
    totalRevenue: { $sum: "$totalAmount" }
 }},
 { $group: {
    _id: null,
    modes: { $push: { mode: "$_id", revenue: "$totalRevenue" } },
    grandTotal: { $sum: "$totalRevenue" }
 }},
 { $unwind: "$modes" },
 { $project: {
    paymentMode: "$modes.mode",
    revenue: "$modes.revenue",
    percentage: { $multiply: [{ $divide: ["$modes.revenue", "$grandTotal"] }, 100] },
    _id: 0
 }},
 { $sort: { revenue: -1 } }
]);
```

REPORT 7: Customer Retention Analysis

MongoDB Aggregation Query:

```
db.orders.aggregate([
 { $group: {
    _id: "$userId",
    totalOrders: { $sum: 1 },
    firstOrder: { $min: "$orderDate" },
```

```
    lastOrder: { $max: "$orderDate" }
 }},
 { $match: { totalOrders: { $gt: 3 } } },
 { $project: {
    userId: "$_id",
    totalOrders: 1,
    repeatPurchaseFrequency: "$totalOrders",
    firstOrder: 1,
    lastOrder: 1,
    _id: 0
 }}
]);
```

ADVANCED: High Value Customers (> ₹1,00,000)

MongoDB Aggregation Query:

```
db.orders.aggregate([
 { $group: {
    _id: "$userId",
    totalSpent: { $sum: "$totalAmount" }
 }},
 { $match: { totalSpent: { $gt: 100000 } } }
]);
```

ADVANCED: Customers inactive for last 60 days

MongoDB Aggregation Query:

```
db.orders.aggregate([
 { $group: {
    _id: "$userId",
    lastOrder: { $max: "$orderDate" }
 }},
 { $match: {
    lastOrder: { $lt: new Date(Date.now() - 60*24*60*60*1000) }
 }}
]);
```

ADVANCED: Real-time Dashboard using $facet

MongoDB Aggregation Query:

```
db.orders.aggregate([
{
 $facet: {
  monthlySales: [
   { $group: {
     _id: { month: { $month: "$orderDate" } },
     revenue: { $sum: "$totalAmount" }
   }}
  ],
  paymentModes: [
   { $group: {
     _id: "$paymentMode",
     revenue: { $sum: "$totalAmount" }
   }}
  ],
  topCustomers: [
   { $group: {
     _id: "$userId",
     spent: { $sum: "$totalAmount" }
   }},
   { $sort: { spent: -1 } },
   { $limit: 5 }
  ]
 }
}
]);
```